

UNIT 8



PERIMETER AND AREA



PERIMETER

“Perimeter” is the measure of the total length of the sides, or line segments of any figure. It is measure of length of the sides, or the boundary of the region. The unit of the perimeter is same as that of length.

Example:

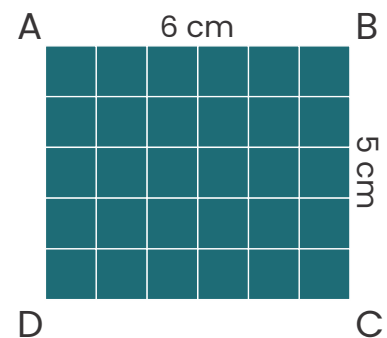
Find the perimeter and area of rectangle with sides of 5 cm and 6 cm.

Solution:

$$\text{Length} = 6 \text{ cm}$$

$$\text{Breadth} = 5 \text{ cm}$$

$$\begin{aligned} \text{Perimeter} &= B + L + B + L \\ &= 5 + 6 + 5 + 6 \\ &= 22 \text{ cm} \end{aligned}$$

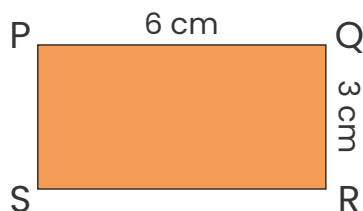


In given rectangle there are total 30 unit squares. So, the area of ABCD = 30 squared cm.

The unit for the perimeter is the same as the length used, i.e., we can use mm, cm, m, and km as the units for measurement of the perimeter.

Examples:

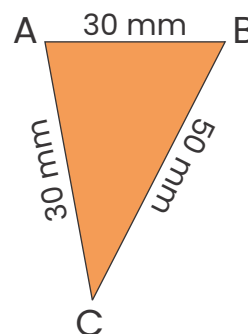
(i)



Here PQRS is a rectangle.

$$\begin{aligned} mPQ &= 6 \text{ cm} \\ mQR &= 3 \text{ cm} \\ mSR &= 6 \text{ cm} \end{aligned}$$

(ii)



Here ABC is a triangle.

Remember 10 mm = 1 cm.

$$\begin{aligned} mAB &= 30 \text{ mm} = 3 \text{ cm} \\ mBC &= 50 \text{ mm} = 5 \text{ cm} \\ mAC &= 40 \text{ mm} = 4 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{mPS} &= + 3 \text{ cm} \\ \text{Perimeter} &= 18 \text{ cm} \end{aligned}$$

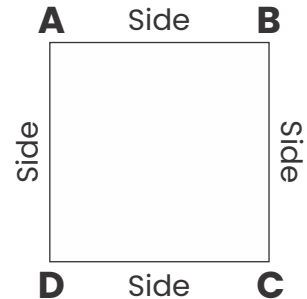
$$\begin{aligned} \text{Perimeter} &= 12 \text{ cm} \\ &= (30 + 50 + 40) \text{ mm} \\ &= 120 \text{ mm} = 12 \text{ cm} \end{aligned}$$

Formulas for the perimeter of square and rectangle:

- (i) We know that a square is a quadrilateral shape in which all its four sides are equal.

Perimeter of the square

$$\begin{aligned} \overline{ABCD} &= \overline{mAB} + \overline{mBC} + \overline{mCD} + \overline{mDA} \\ &= \text{side} + \text{side} + \text{side} + \text{side} \\ &= 4 \times \text{side} \\ &= 4 \times (\text{length of a side of a square}) \end{aligned}$$

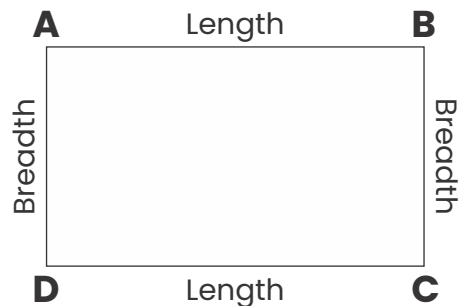


Thus, formula for perimeter of a square figure = $4 \times \text{length of a side}$

- (ii) Draw a rectangle ABCD. Measure its sides.
Then find its perimeter as follows:

Perimeter of the rectangle ABCD

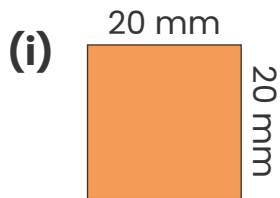
$$\begin{aligned} \overline{mABCD} &= \overline{mAB} + \overline{mBC} + \overline{mCD} + \overline{mDA} \\ &= \text{length} + \text{breadth} + \text{length} + \text{breadth} \\ &= 2 \times \text{length} + 2 \times \text{breadth} \\ &= 2 \times (\text{length} + \text{breadth}) \end{aligned}$$



Thus, formula for perimeter of a rectangle = $2 \times (\text{length} + \text{breadth})$

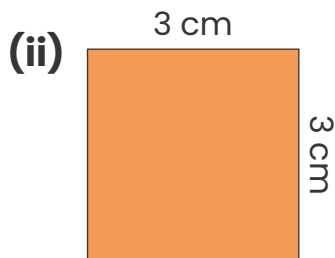
EXERCISE 8.1

Q1. Find the perimeter and write the unit of perimeter for each of the following.



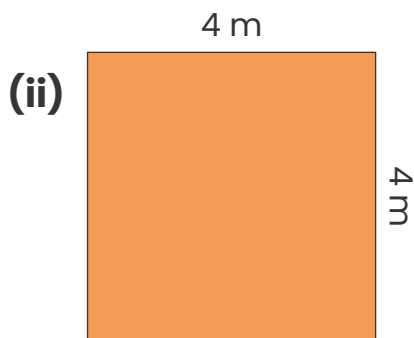
(a) Here perimeter _____.

(b) Unit of perimeter _____.



(a) Here perimeter _____.

(b) Unit of perimeter _____.

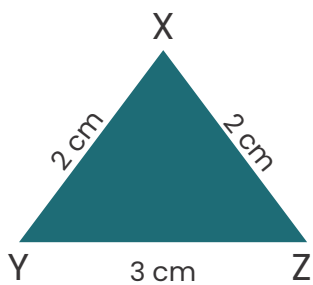


(a) Here perimeter _____.

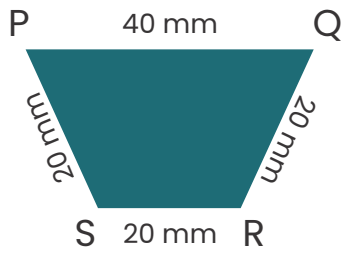
(b) Unit of perimeter _____.

Q2. Find the perimeter of each of the following figures:

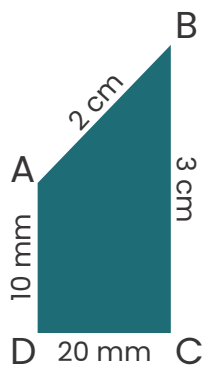
(1)



(2)



(3)



Q3. Find the perimeter of each of the following rectangles.

$L = 3\text{cm}$, $B = 2\text{cm}$

Solution:

$L = 5\text{cm}$, $B = 1\text{cm}$

Solution:

$L = 4\text{cm}$, $B = 3\text{cm}$

Solution:

Q4. Find perimeter of a square whose sides are given below:

4cm

Solution:

6cm

Solution:

7.5cm

Solution:

AREA

“Area” is the measure of the region of a closed figure. We can find the area of a closed figure by fitting units squares.

Example:

Find area of a rectangle whose length is 5 cm and breadth is 2 cm.

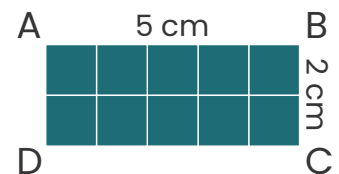
Solution:

Here length = 5 cm
breadth = 2 cm

Area of a rectangle ABCD:

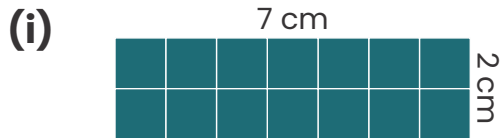
$$= \text{Length} \times \text{Breadth}$$

$$= 5 \times 2 = 10 \text{ sq. cm}$$



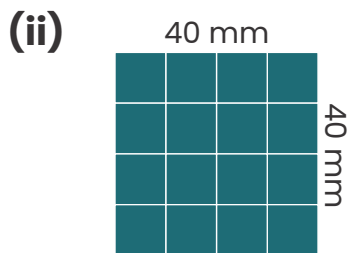
EXERCISE 8.2

Q1. Find the area by counting small unit squares and write the unit of area for each of the following.



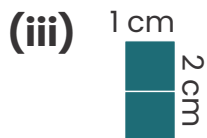
(a) Area _____ squares.

(b) Unit of area _____.



(a) Area _____ squares.

(b) Unit of area _____.



(a) Area _____ squares.

(b) Unit of area _____.

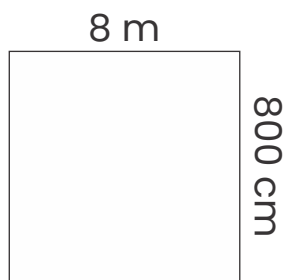


(a) Area _____ squares.

(b) Unit of area _____.

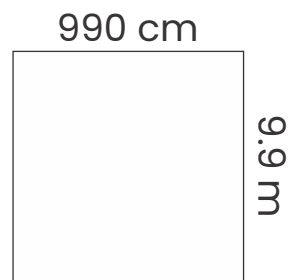
Q2. Find the area of the following shapes by using formula.

(i)



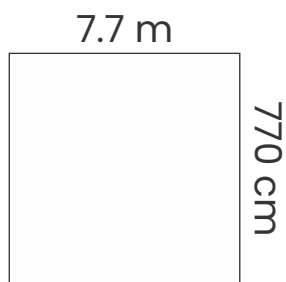
Solution:

(ii)



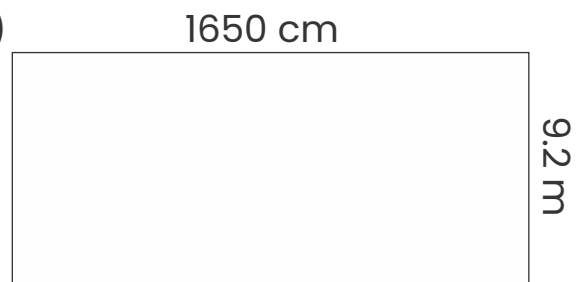
Solution:

(iii)



Solution:

(iv)



Solution:

Q3. Find the area and perimeter of each of the following rectangles.

$$L = 4\text{cm } B = 2\text{cm}$$

Solution:

$$L = 5\text{cm } B = 2\text{cm}$$

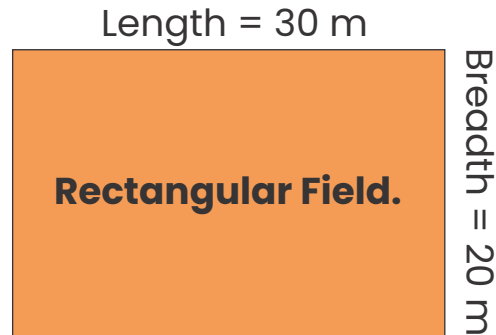
Solution:

$$L = 6\text{cm } B = 4\text{cm}$$

Solution:

Example 1: Length of a rectangular field is 30 m and its breadth is 20 m. Find the perimeter of the rectangular field.

Solution:



Length of the given rectangular field = 30 m

Breadth of the given rectangular field = 20 m

Formula: Perimeter of the given rectangular field is

$$\begin{aligned}
 & 2 \times (\text{Length} + \text{Breadth}) \\
 &= 2 \times (30 \text{ m} + 20 \text{ m}) \\
 &= 2 \times 50 \text{ m} \\
 &= 100 \text{ m}
 \end{aligned}$$

Example 2: A square ground has each side 36 m. What distance will a boy cover by cycling around the ground three times?

Solution:

The side of the square ground is 36 m each.

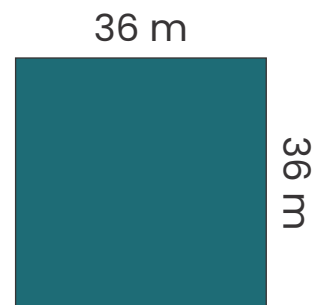
Therefore perimeter of the ground is: $4 \times \text{side}$

$$= 4 \times 36 \text{ m}$$

$$= 144 \text{ m}$$

In one cycle time boy will cover 144 m distance.

In three times the boy will cover $144 \times 3 = 432 \text{ m}$



$$\begin{array}{r}
 144 \\
 \times 3 \\
 \hline
 432
 \end{array}$$

Example 3: The perimeter of a rectangular park is 320 m. Its length is 70 m. Find the breadth.

Solution:

$$\text{Perimeter} = 320 \text{ m}$$

$$\text{Length} = 70 \text{ m}$$

$$\text{Perimeter} = 2 \times (\text{length} + \text{breadth})$$

$$\text{Breadth} = \frac{\text{Perimeter}}{2} - \text{length}$$

$$= \left(\frac{320}{2} \right) - 70 = \frac{160}{1} - 70$$

$$= 160 - 70 = 90 \text{ m}$$

Example 4: The perimeter of a square shape is 280 cm. Find the length of each side.

Solution:

$$\text{Perimeter} = 280 \text{ m}$$

$$\text{Length of each side} = \frac{\text{Perimeter}}{4}$$

$$= \frac{280}{4} = 70 = 70 \text{ cm}$$

EXERCISE 8.3

Q1. A rectangular garden is 120 m long and 80 m wide. What is the perimeter of the garden?

Solution:

Q2. A square classroom is 10 m wide. What is the area of the classroom?

Solution:

Q3. A square painting measures 50 cm on each side. How much wooden frame is required to cover it on all sides?

Solution:

Q4. The length and width of a rectangular playground are 150 m and 100 m, respectively. Calculate the area and perimeter of the playground.

Solution:
